Run report: H7-subgroup-run-20261018T2209Z

Descriptive record of one recorded run of examples/decomposition_coverage.rs, mode subgroup (Addendum H7) under `research/weil_descent_angles_20261018/PROTOCOL.md` (Addendum A and Amendments B–G and Addendum H, each committed before the run it governs). It lists what ran, on what host, and what the tool recorded, file by file. Interpretation is in the study's `REPORT.md`. Costs are counted operations (F_p multiplications and inversions, group operations, native operations), never wall-clock; the only times are the tool's own seconds. Isolation: none earned (a shared cloud container, or a RunPod CPU pod whose lscpu is in host.txt); this affects only the seconds, since every recorded value is an exact count or a counted operation total.

env.txt

```
commit 6083c58aa45e04e4d3194689b3b3e6ef5c7be115
rustc 1.97.0 (2d8144b78 2026-07-07)
4
Sun Oct 18 22:09:13 UTC 2026
host: this session's cloud container (L0)
cmd: decomposition_coverage <dir> subgroup
```

checks.txt

```
H7 binary n=2: ell-subgroup coverage ratio subfield/null mean 1.864 (s.e. 0.155, 21 curves)
H7 binary n=4: ell-subgroup coverage ratio subfield/null mean 2.069 (s.e. 0.714, 9 curves)
H7 odd_generic n=2: ell-subgroup coverage ratio subfield/null mean 1.168 (s.e. 0.103, 12 curves)
H7 odd_generic n=4: ell-subgroup coverage ratio subfield/null mean 1.332 (s.e. 0.533, 9 curves)
H7 odd_split_twist n=2: ell-subgroup coverage ratio subfield/null mean 1.180 (s.e. 0.550, 12 curves)
H7 odd_split_twist n=4: ell-subgroup coverage ratio subfield/null mean 0.402 (s.e. 0.150, 9 curves)
H7 odd_split_twist_matched n=2: ell-subgroup coverage ratio subfield/null mean 2.393 (s.e. 0.214, 12 curves)
H7 odd_split_twist_matched n=4: ell-subgroup coverage ratio subfield/null mean 0.742 (s.e. 0.130, 9 curves)
```

subgroup.csv (first 12 of 415 rows)

kind	n	q	curve	order	ell	cofactor	layer	layer_size	F_sub	F_null	cov_sub	cov_null	ratio
binary	2	16	0	276	23	12	E	276	15	16	0.322464	0.409420	0.78761
binary	2	16	0	276	23	12	2E	138	15	16	0.644928	0.514493	1.25352
binary	2	16	0	276	23	12	4E	69	15	16	0.681159	0.449275	1.51613
binary	2	16	0	276	23	12	ell	23	15	16	0.652174	0.565217	1.15385
binary	2	16	1	242	11	22	E	242	17	18	0.442149	0.413223	1.07000
binary	2	16	1	242	11	22	2E	121	17	18	0.884298	0.438017	2.01887
binary	2	16	1	242	11	22	ell	11	17	18	0.818182	0.818182	1.00000
binary	2	16	2	270	5	54	E	270	17	12	0.403704	0.270370	1.49315
binary	2	16	2	270	5	54	2E	135	17	12	0.807407	0.303704	2.65854
binary	2	16	2	270	5	54	ell	5	17	12	1.000000	0.600000	1.66667
binary	2	32	0	1024	2	512	E	1024	31	28	0.356445	0.338867	1.05187
binary	2	32	0	1024	2	512	2E	512	31	28	0.712891	0.431641	1.65158

Graph

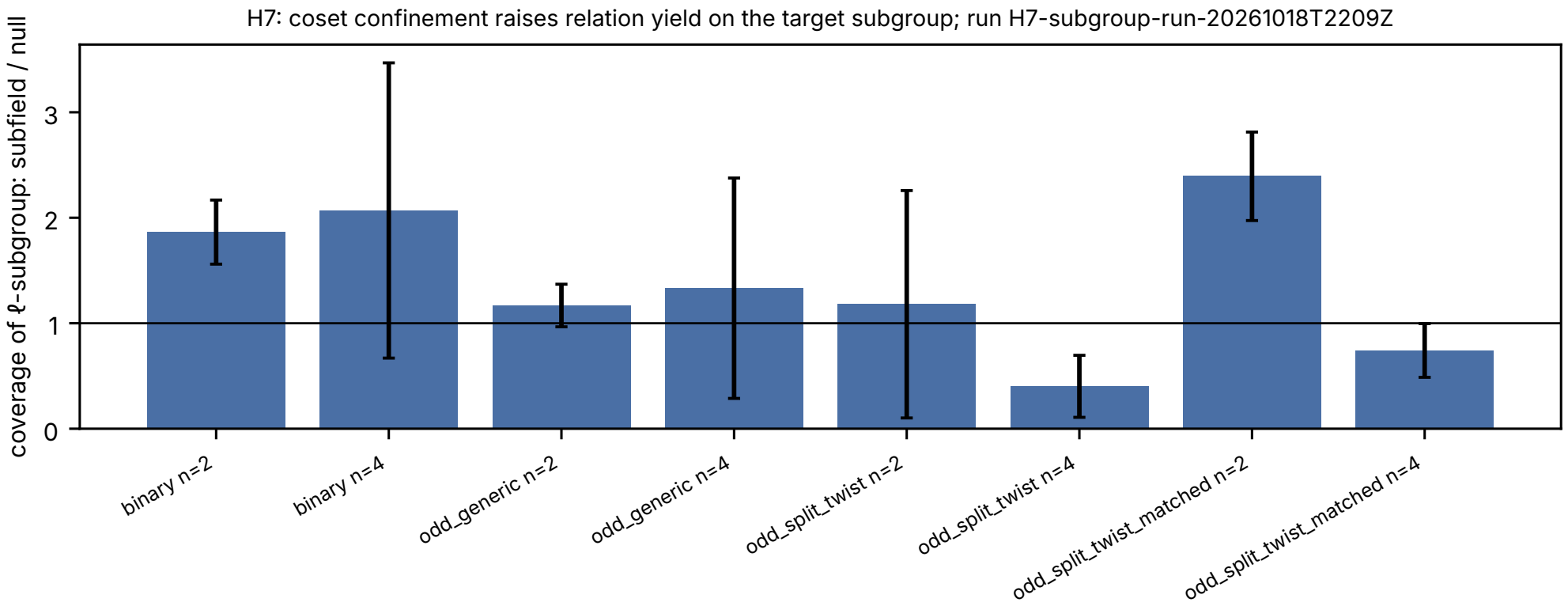


Figure 1: H7: coverage of the l-subgroup, subfield over null, mean and 1.96 s.e. per cell.